MODULE 19 · DOMAIN 3 - APPLICATION DEVELOPMENT

Genie Agents and multi-agent data access

Questions over tables, answered with SQL, and three ways to bring that ability into a multi-agent system.

SECTION 3 of 6 EXAM WEIGHT 30% MODULE 19 of 40

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Genie is the analyst down the corridor. **Ask it a question about your tables, and it writes the SQL, runs it, and tells you the answer.**

Genie, or document retrieval?

GENIE AGENT

Questions about tables

- Totals, counts, trends, comparisons
- Writes and runs SQL
- Unity Catalog tables and views only
- Cannot read PDFs or document files

RETRIEVAL OR KNOWLEDGE ASSISTANT

Questions about documents

- Policies, manuals, contracts
- Finds and cites passages
- Chunks in an AI Search index
- Cannot total a column

THE TIEBREAKER

If the answer is a number computed from rows, use Genie. If it is a passage from a document, use retrieval.

What makes a Genie Agent answer correctly

INGREDIENT	WHY IT HELPS
Only the right tables	Genie can query only tables you add. Fewer, relevant ones
Column descriptions	Explain cryptic names such as amt_net_2
Instructions	Business definitions, such as fiscal quarters or active customer
Example SQL queries	Show how questions map to correct SQL
Join instructions	How tables relate, so joins are right
Benchmarks	Test questions with known answers, rerun after changes

ON THE EXAM

When Genie gets a business definition wrong, add an instruction or an example query. The model is fine. Its knowledge of your business is not.

Asking Genie a question from code

01
Start
POST start-conversation with the question.

02
Poll
Check the message status every 1 to 5 seconds.

03
Read
Attachments hold the SQL and description.

04
Fetch
Get the query result for the table of answers.

05
Follow up
Ask again in the same conversation.

ON THE EXAM

Genie works asynchronously. Starting a conversation returns immediately; the answer arrives later, so you poll.

The operational rules

RULE	DETAIL
Poll until conclusive	COMPLETED, FAILED or CANCELLED. Give up after about 10 minutes
Retry with backoff	The API does not retry failed requests for you
One conversation per session	Reusing one thread for many sessions harms accuracy
Service principal	When browser sign-in is not possible
Trusted answers	A query.parameters object shows a trusted asset was used

ON THE EXAM

Useful content, such as the generated SQL, appears in the attachments before COMPLETED, while the query is still running.

Three ways to bring Genie in

01

Supervisor Agent

Add the Genie Agent as a subagent. No code.

02

Managed MCP

Genie as a tool for a custom agent.

03

Conversation API

Call it from your own code, or the LangChain Genie wrapper.

ON THE EXAM

All three reach the same Genie Agent. Choose by how much control you need, least first.

Supervisor subagent, or a tool in custom code?

SUPERVISOR AGENT SUBAGENT

Configure

- Existing Genie and Knowledge Assistants
- Routing described in words
- No code to maintain
- Tuned with expert feedback

MCP TOOL OR API IN CUSTOM CODE

Build

- A custom agent already exists
- Genie is one tool among several
- Custom logic around the result
- Full control of calls and retries

THE TIEBREAKER

Coordinating existing agents: Supervisor Agent. Adding Genie to an agent you already wrote: the managed MCP server.

Whose data does Genie see?

- Genie evaluates data access with **each end user's own Unity Catalog permissions**.
- Row filters and column masks on the tables **still apply** inside Genie.
- Inside a multi-agent system, each user still sees only the rows they may see.
- If an app calls Genie as a **service principal**, it sees what the principal may see.
- A prompt telling Genie to hide rows is not a security control.

> ON THE EXAM
>
> Restrict data in Unity Catalog, not in the prompt. Genie then enforces it for every user, in every agent that calls it.

Chat mode, or Agent mode?

CHAT MODE

One question, one answer

- The default
- Generates and runs SQL
- Table, text, maybe a chart
- Fast, for direct questions

AGENT MODE

A research plan

- Plans, runs several queries, iterates
- Report with citations and tables
- Streams results as events
- For open, multi-step questions

THE TIEBREAKER

A direct question wants chat mode. Why did sales fall in the north, and what drove it, wants agent mode.

Harbourline's claims analytics

01
Genie Agent
Claims tables, fiscal-year instructions.

02
Security
Row filter: adjusters see their region.

03
Tool
Added to the claims agent via managed MCP.

04
Documents
Policy wording stays in AI Search.

05
Benchmarks
Forty questions rerun after each change.

ON THE EXAM

Numbers from Genie, passages from retrieval, and security in Unity Catalog, not in any prompt.

> A retailer's Genie Agent answers sales questions. The company's financial year starts in February, so its Q1 runs from February to April.
>
> When staff ask about Q1 sales, Genie returns January to March figures.

What should the Genie Agent's author change?

A. Add more tables to the Genie Agent so it has more data

B. Switch to a larger SQL warehouse so queries run faster

C. Replace Genie with a vector index built over the sales tables

D. Add an instruction defining the company's fiscal quarters

◆ PAUSE AND DECIDE

What should the Genie Agent's author change?

A More tables

The data is fine. The definition of Q1 is wrong.

B Larger warehouse

Faster wrong answers.

C Vector index over tables

Cannot compute quarterly totals at all.

D Fiscal-quarter instruction

Teaches Genie the business definition it lacked.

THE SIGNAL

Calendar quarters instead of fiscal ones. Genie is missing a business definition, so give it one as an instruction.

> An app sends questions to a Genie Agent with the conversation API. It calls start-conversation and immediately reads the response for the answer.
>
> The answer is usually missing, although the Genie Agent works fine in the user interface.

What should the developer change?

A Call start-conversation again until an answer appears in it

B Poll the message status until COMPLETED, then fetch the result

C Use a larger SQL warehouse so the answer returns instantly

D Reuse one shared conversation for every user to save time

◆ PAUSE AND DECIDE

What should the developer change?

A Start again until it appears

Each call starts a new conversation. It never waits.

B Poll, then fetch the result

Genie answers asynchronously. Wait for COMPLETED.

C Larger warehouse

Faster, but still not instant. Polling is still needed.

D One shared conversation

Harms accuracy, and does not make answers arrive sooner.

THE SIGNAL

Reads the response immediately. Genie is asynchronous: start returns at once, and the answer must be polled for.

> To save effort, an app sends every user's questions to a Genie Agent through one long-running conversation thread.
>
> Over time, answers start mixing up context from different users' questions.

What should the developer change?

A Start a new conversation for each user session

B Poll the API more often, every half a second instead

C Add every table in the catalog to the Genie Agent

D Switch the Genie Agent from chat mode to agent mode

◆ PAUSE AND DECIDE

What should the developer change?

A New conversation per session

Each session gets its own context. Databricks' own advice.

B Poll more often

Faster polling does not separate contexts.

C Every table in the catalog

More confusion, not less. Fewer relevant tables work better.

D Agent mode

Still one shared thread, with the same mixing.

THE SIGNAL

One thread for everyone, and context mixing. A conversation carries context, so each session needs its own.

> A multi-agent assistant includes a Genie Agent over the sales tables. Regional managers must see only their own region's sales.
>
> The sales table already has a Unity Catalog row filter by region, and users call the assistant as themselves.

How is the regional restriction best enforced?

A Tell the supervisor's prompt to answer only about the user's region

B Give the app a service principal with access to every region

C Rely on Genie applying each end user's own Unity Catalog permissions

D Create a separate copy of the sales table for every region

◆ PAUSE AND DECIDE

How is the regional restriction best enforced?

A Prompt the supervisor
A prompt is not a security control.

B All-region service principal
Everyone would see every region.

C Genie uses each user's permissions
The existing row filter applies automatically.

D A table copy per region
Duplicated data, and the filter already exists.

THE SIGNAL

A row filter already exists, and users call as themselves. Genie applies each end user's permissions, so the filter just works.

What to carry forward

- Genie answers **questions about tables** with SQL. Documents need retrieval.
- Accuracy: right tables, descriptions, **instructions**, example SQL, benchmarks.
- Conversation API: start, **poll until COMPLETED**, fetch result. **New conversation per session**.
- In multi-agent systems: Supervisor subagent, managed MCP tool, or the API.
- Genie uses each **end user's own permissions**. Security lives in Unity Catalog.

> ON THE EXAM
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> Next: module 20 separates evaluation from monitoring, and chooses chunking from evidence.